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Indian Institute of Technology Hyderabad

Mathematical Foundations of Data Science (Probability)

Lecture 05

Karthik P. N.

Assistant Professor, Department of AI

Email: pnkarthik@ai.iith.ac.in

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Independence of Two Discrete Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Assume that all random variables appearing below are defined with respect to $\mathcal{F}$.

Theorem 1 (Independence of Jointly Discrete Random Variables)

Let X and Y be jointly discrete random variables. The following statements are equivalent.

1. X is independent of Y .
2. For all $x, y \in \mathbb{R}$, the event $\{X = x\}$ is independent of the event $\{Y = y\}$.
3. $p_{X,Y}(x, y) = p_X(x) \cdot p_Y(y)$ for all $x, y \in \mathbb{R}$.
4. $p_{X|\{Y=y\}}(x) = p_X(x)$ for all $x, y \in \mathbb{R}$ such that the conditional PMF is well-defined.

Remark:

- The four statements above equivalent means

$$1 \implies 2 \implies 3 \implies 4 \implies 1.$$

Jointly Continuous Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ be random variables defined with respect to $\mathcal{F}$.

Definition (Jointly Continuous Random Variables)

The random variables X and Y are said to be **jointly continuous** if there exists a non-negative function $f_{X,Y} : \mathbb{R}^2 \rightarrow [0, +\infty)$ such that

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) \, dv \, du \quad \forall x, y \in \mathbb{R}.$$

The function $f_{X,Y}$ is called the **joint PDF** of X and Y .

Properties of Joint PDF

Theorem 2 (Properties of Joint PDF)

Let X and Y be **jointly continuous** random variables with joint PDF $f_{X,Y}$. Then:

$$1. \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(u, v) \, dv \, du = 1.$$

2. For any $u \in \mathbb{R}$,

$$\int_{-\infty}^{\infty} f_{X,Y}(u, v) \, dv = f_X(u).$$

(This says if X and Y are jointly continuous RVs, then X is a continuous RV.)

3. For any $v \in \mathbb{R}$,

$$\int_{-\infty}^{\infty} f_{X,Y}(u, v) \, du = f_Y(v).$$

(This says if X and Y are jointly continuous RVs, then Y is a continuous RV.)

Conditional PDF

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ be **jointly continuous** random variables defined with respect to $\mathcal{F}$.

Conditional CDF of X conditioned on $\{Y = y\}$: $\mathbb{P}(\{X \leq x\} | \{Y = y\})$.

However, this conditional probability is not defined because $\mathbb{P}(\{Y = y\}) = 0$.

Remedy:

- Fix $y \in \mathbb{R}$ and $\varepsilon > 0$ such that $\mathbb{P}(\{Y \in (y - \varepsilon, y + \varepsilon)\}) > 0$
- Define conditional probability with respect to the event $\{Y \in (y - \varepsilon, y + \varepsilon)\}$, and let $\varepsilon \downarrow 0$

Conditional PDF

$$\begin{aligned}
 \mathbb{P}(\{X \leq x\} | \{Y \in (y - \varepsilon, y + \varepsilon)\}) &= \frac{\mathbb{P}(\{X \leq x\} \cap \{Y \in (y - \varepsilon, y + \varepsilon)\})}{\mathbb{P}(\{Y \in (y - \varepsilon, y + \varepsilon)\})} \\
 &= \frac{\int_{-\infty}^x \int_{y-\varepsilon}^{y+\varepsilon} f_{X,Y}(u, v) \, dv \, du}{\int_{y-\varepsilon}^{y+\varepsilon} f_Y(v) \, dv} \\
 &\approx \frac{\int_{-\infty}^x f_{X,Y}(u, y) \, du \cdot 2\varepsilon}{f_Y(y) \cdot 2\varepsilon} \\
 &= \int_{-\infty}^x \underbrace{\frac{f_{X,Y}(u, y)}{f_Y(y)}}_{\text{conditional PDF}} \, du
 \end{aligned}$$

Conditional CDF for Jointly Continuous RVs

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ be **jointly continuous** random variables defined with respect to $\mathcal{F}$.

Definition (Conditional CDF for Jointly Continuous Random Variables)

The **conditional CDF** of X , conditioned on the event $\{Y = y\}$, is the function $F_{X|Y=y} : \mathbb{R} \rightarrow [0, 1]$ defined as

$$F_{X|Y=y}(x) = \int_{-\infty}^x \frac{f_{X,Y}(u, y)}{f_Y(y)} du, \quad x \in \mathbb{R},$$

defined for all $y \in \mathbb{R}$ such that $f_Y(y) > 0$.

Conditional PDF for Jointly Continuous RVs

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ be **jointly continuous** random variables.

Definition (Conditional PDF for Jointly Continuous Random Variables)

The **conditional CDF** of X , conditioned on the event $\{Y = y\}$, is the function $f_{X|\{Y=y\}} : \mathbb{R} \rightarrow [0, +\infty)$ given by

$$f_{X|\{Y=y\}}(x) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \quad x \in \mathbb{R},$$

defined for all $y \in \mathbb{R}$ such that $f_Y(y) > 0$.

Independence and Joint Continuity

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Theorem 3 (Joint Continuity and Independence)

Two random variables X and Y which are **jointly continuous** are independent if and only if

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) \quad \forall x, y \in \mathbb{R}.$$

Remark:

- $X \perp\!\!\!\perp Y \iff f_{X|\{Y=y\}} = f_X$ for all y such that $f_Y(y) > 0$

Conditional PMF and Conditional PDF – 1

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let X and Y be **jointly discrete** with $\mathbb{P}(\{(X, Y) \in \mathcal{E}\}) = 1$ for some discrete $\mathcal{E} \subset \mathbb{R}^2$.

- Define the projection sets

$$\mathcal{X} := \{x \in \mathbb{R} : (x, y) \in \mathcal{E} \text{ for some } y \in \mathbb{R}\}, \quad \mathcal{Y} := \{y \in \mathbb{R} : (x, y) \in \mathcal{E} \text{ for some } x \in \mathbb{R}\}.$$

- The conditional PMF of X , conditioned on the event $\{Y = y\}$, is defined as

$$p_{X|\{Y=y\}}(x) = \frac{p_{X,Y}(x, y)}{p_Y(y)}, \quad x \in \mathbb{R}, \quad p_Y(y) > 0.$$

- For any “nice” set $B \subseteq \mathbb{R}$,

$$\mathbb{P}(\{X \in B\} | Y = y) = \sum_{x \in \mathcal{X} \cap B} p_{X|\{Y=y\}}(x).$$

- Along similar lines,

$$\mathbb{P}(\{Y \in B\} | X = x) = \sum_{y \in \mathcal{Y} \cap B} p_{Y|\{X=x\}}(y),$$

defined for all x such that $p_X(x) > 0$

Conditional PMF and Conditional PDF – 2

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let X and Y be **jointly continuous** with joint PDF $f_{X,Y}$.

- The conditional PDF of X , conditioned on the event $\{Y = y\}$, is defined as

$$f_{X|\{Y=y\}}(x) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \quad x \in \mathbb{R}, f_Y(y) > 0.$$

- For any “nice” set $B \subseteq \mathbb{R}$,

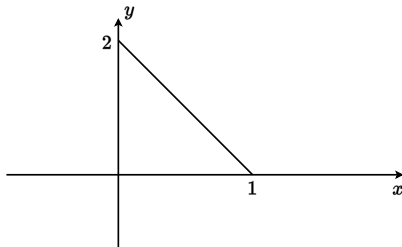
$$\mathbb{P}(\{X \in B\} | \{Y = y\}) = \int_B f_{X|\{Y=y\}}(u) \, du.$$

- Along similar lines,

$$\mathbb{P}(\{Y \in B\} | \{X = x\}) = \int_B f_{Y|\{X=x\}}(v) \, dv,$$

defined for all x such that $f_X(x) > 0$

Problem-01



Let (X, Y) be uniformly distributed on the right-triangle.

- Compute the marginal PDFs of X and Y .
- Compute the conditional PDF of X conditioned on $\{Y = y\}$ for various values of y .
- Argue if X and Y are independent.

Problem-02

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Suppose X is a discrete random variable defined with respect to $\mathcal{F}$ and having the PMF

$$p_X(100) = 0.65, \quad p_X(-10) = 0.35.$$

Let $Y = \max\{X, 0\}$, i.e., $Y(\omega) = \max\{X(\omega), 0\}$ for each $\omega \in \Omega$.

1. Show that Y is a discrete random variable with respect to $\mathcal{F}$.
2. Sketch the CDFs of X and Y .

Problem-03

Suppose that X and Y are jointly discrete, integer-valued random variables, with the joint PMF

$$p_{X,Y}(m, n) = \frac{\lambda^n e^{-2\lambda}}{m! (n - m)!}, \quad 0 \leq m \leq n < +\infty, \quad m, n \in \{0, 1, 2, \dots\}.$$

1. Determine a discrete set $\mathcal{E} \subset \mathbb{R}^2$ such that $\mathbb{P}((X, Y) \in \mathcal{E}) = 1$.
2. Based on the result above, determine

$$\mathcal{X} := \{x \in \mathbb{R} : \exists y \in \mathbb{R} \text{ such that } (x, y) \in \mathcal{E}\}, \quad \mathcal{Y} := \{y \in \mathbb{R} : \exists x \in \mathbb{R} \text{ such that } (x, y) \in \mathcal{E}\}.$$

3. Determine the marginal PMFs of X and Y .
4. Determine the conditional PMF of X conditioned on the event $\{Y = n\}$ for some fixed $n \in \{0, 1, 2, \dots\}$.
5. Are X and Y independent? Justify.

Problem-04

Suppose that X and Y are jointly continuous with the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cy, & -1 \leq x \leq 1, \ 0 \leq y \leq |x|, \\ 0, & \text{otherwise.} \end{cases}$$

1. Determine the constant c .
2. Are X and Y independent?
3. Evaluate $\mathbb{P}(\{X \geq Y + 0.5\})$.
4. Compute the conditional PDF of X , conditioned on the event $\{Y > 0.5\}$.
Using this, evaluate $\mathbb{P}(\{X > 0.75\}|\{Y > 0.5\})$.